

The Changing Nature of Policing

The role of the police has evolved over time, and they are now being viewed as the primary public service in many situations. This shift in perception is a result of their versatile functions and responsibilities. As societies face new and more complex challenges, police workloads increasingly focus on managing risks of harm to vulnerable people. To best inform this expanded scope of duties we sought to explore the changing nature and temporal and spatial dependencies related to both current and historical police related demand across the Bradford region. By recognising and analysing these patterns, we hope to inform police and their partners of the best ways to allocate resources and effectively respond to the needs of the communities they serve.

The Data

Police incident data, is when people called West Yorkshire Police (WYP) for police service. In this study we analysed incidents graded emergency or priority and physically attended by police officers.

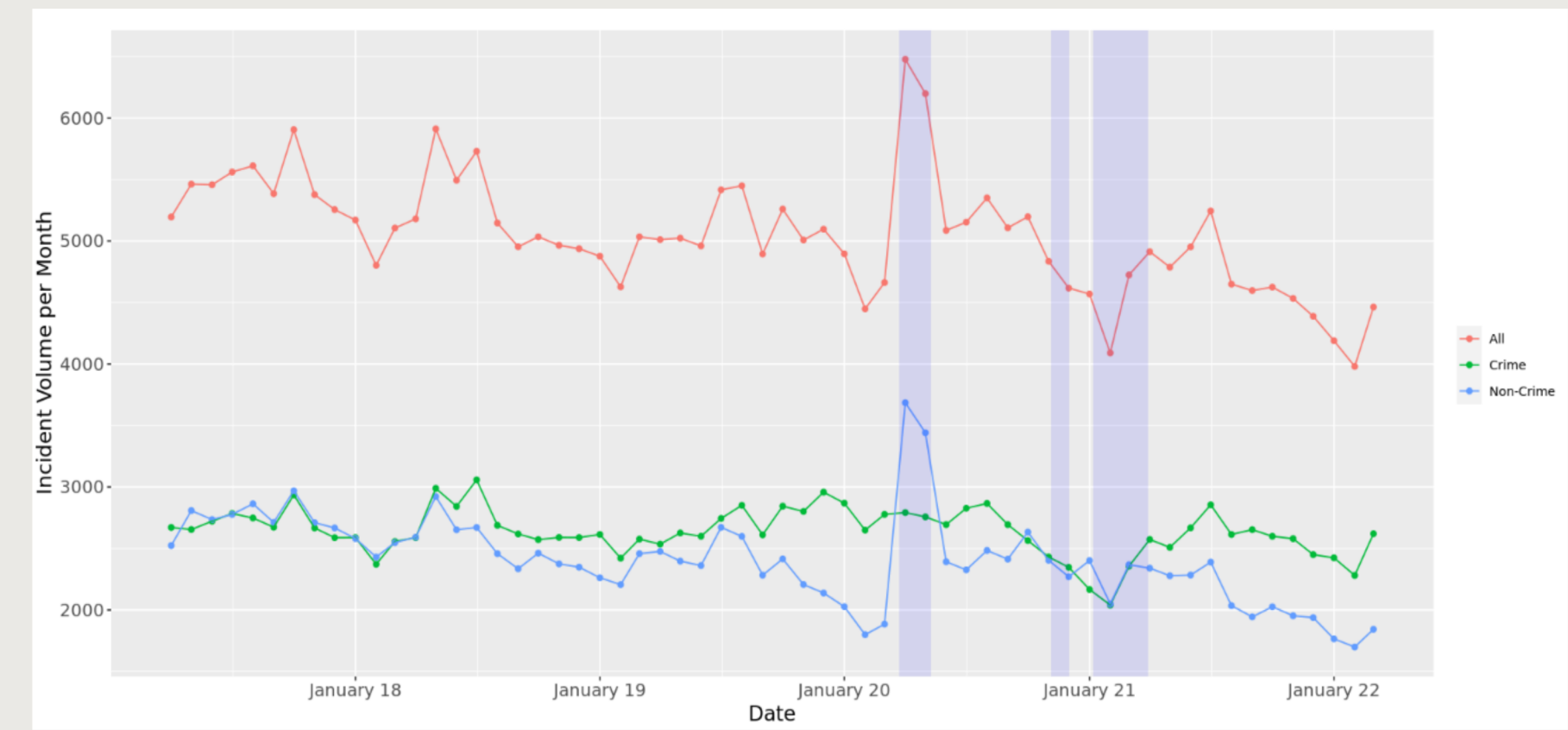


Figure 1 Incident volume plot for attended incidents graded emergency and priority in the five year study period. Trend for all incidents (red), crime related (green) and non crime related (blue). Blue shaded region denotes national lockdown restrictions.

- Study Period: **April 2017 to March 2022**
- WYP recieved **330k** calls for service
- **89** types, **21** subtypes and **39** qualifiers
- 48% of calls were not related to crime.

Quantifying Vulnerability-Related Demand

- There are two measures of demand: The **volume** of calls recieved over a given time period and **duration** of time associated with responding to each of those calls.
- We combine these measures to calculate **accumulated** response time - measure of the total amount of time spent responding to particular incidents in each neighbourhood.
- Developed an (explainable, equitable, extensible) expert consensus mehcanism for identifying incidents that may be indicative of vulnerability

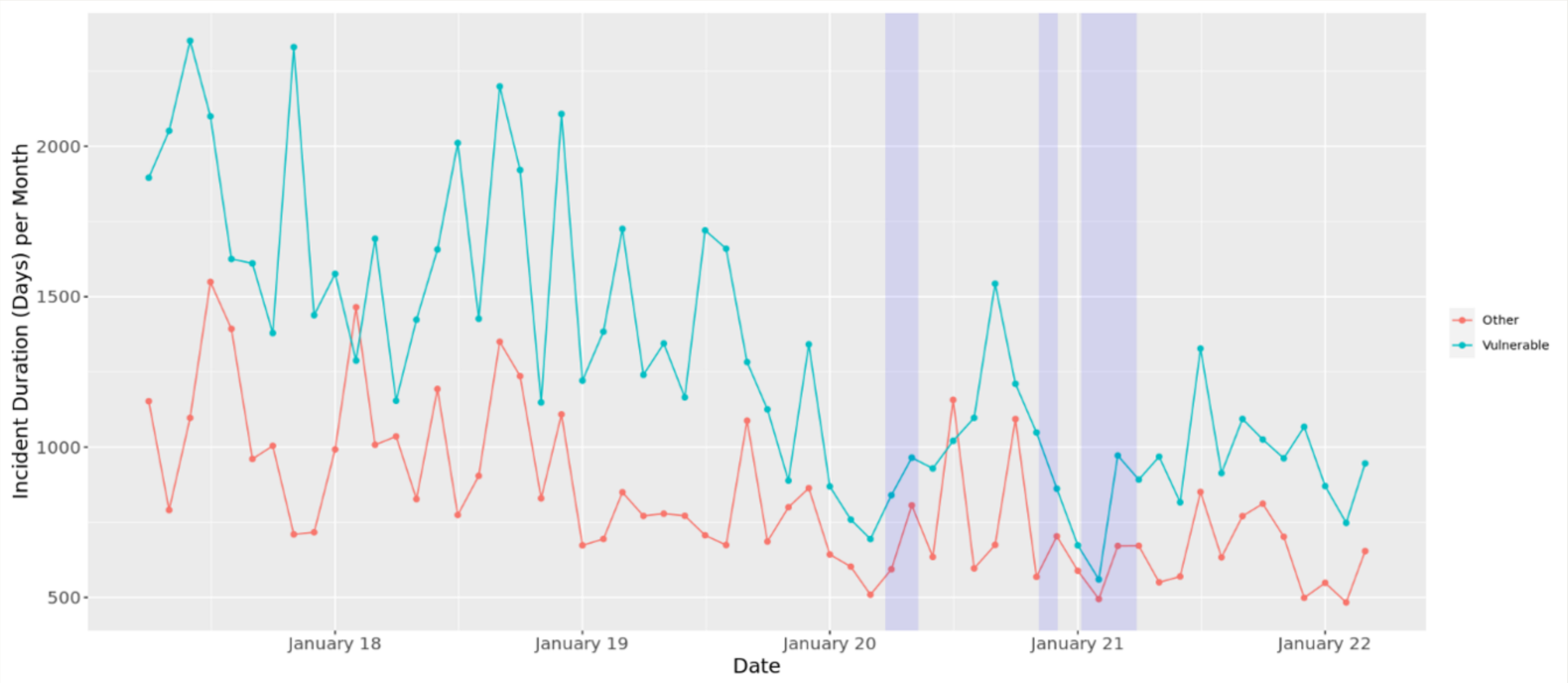


Figure 2 Temporal trends for Vulnerability-Related incident durations across the five year study period, considering attended incidents of emergency and priority grading. Blue shaded region denotes national lockdown restrictions.

The graph above and the table below shows that both when considering volume and duration, our estimates indicate that a significant amount of time is spent dealing with vulnerability related incidents.

	Volume	Duration (Days)	Volume (%)	Duration (%)
Vulnerability Related	162013	78143.4	54	61
Other	141002	49526.6	47	39

Table 2 Table showing the amount and percentage for vulnerability related and other incidents. Considering attended incidents given grading emergency and priority.

Key Findings

- Analysing attended emergency and priority calls - 48% of calls were non-crime related, and 41% of time spent responding was non-crime related.**
- Identifying vulnerabilities is a complex task**
Consolidating experts provided an equitable, explainable and extendable means to identify vulnerability-related incidents - nevertheless, there are distinct differences in expert opinion regarding what constitutes a vulnerability.
- Analysing attended emergency and priority calls - 54% of calls were potentially vulnerability related, and 61% of time spent responding was potentially vulnerability related.**
- Analysing attended potentially vulnerability related emergency and priority calls - 52% of calls were related to non-crime, and 37% of time spent responding was non-crime.**
- There are high levels of spatial concentration in vulnerability related demand across the region.**
With some LSOA's experiencing up to125x higher demand than the lowest.

Estimating Vulnerability Related Demand

To try and understand which incidents might be indicative of vulnerability we asked a panel of 3 experts to examine a range of markers collected for each incident: "***Would you consider this type/subtype/qualifier to be indicative of a vulnerability?***"

Conensus Mechanism: Incident classified as vulnerability-related, if the majority of experts consider type, subtype or qualifier(s) to be vulnerability-related

Type	Sub Type	Qualifiers	Vulnerability Related
Sexual Offence	Victim aged 16+	Mental Health, Domestic	Yes
Burglary - residential	Niche created occurrence	Domestic abuse	Yes
Burglary - residential	Niche created occurrence	No app ho qualifier	No

Table 1 Illustrating how the Consensus Mechanism Coding works, when determining whether or not an incident is vulnerability-related.

Vulnerability-Related Spatial Demand

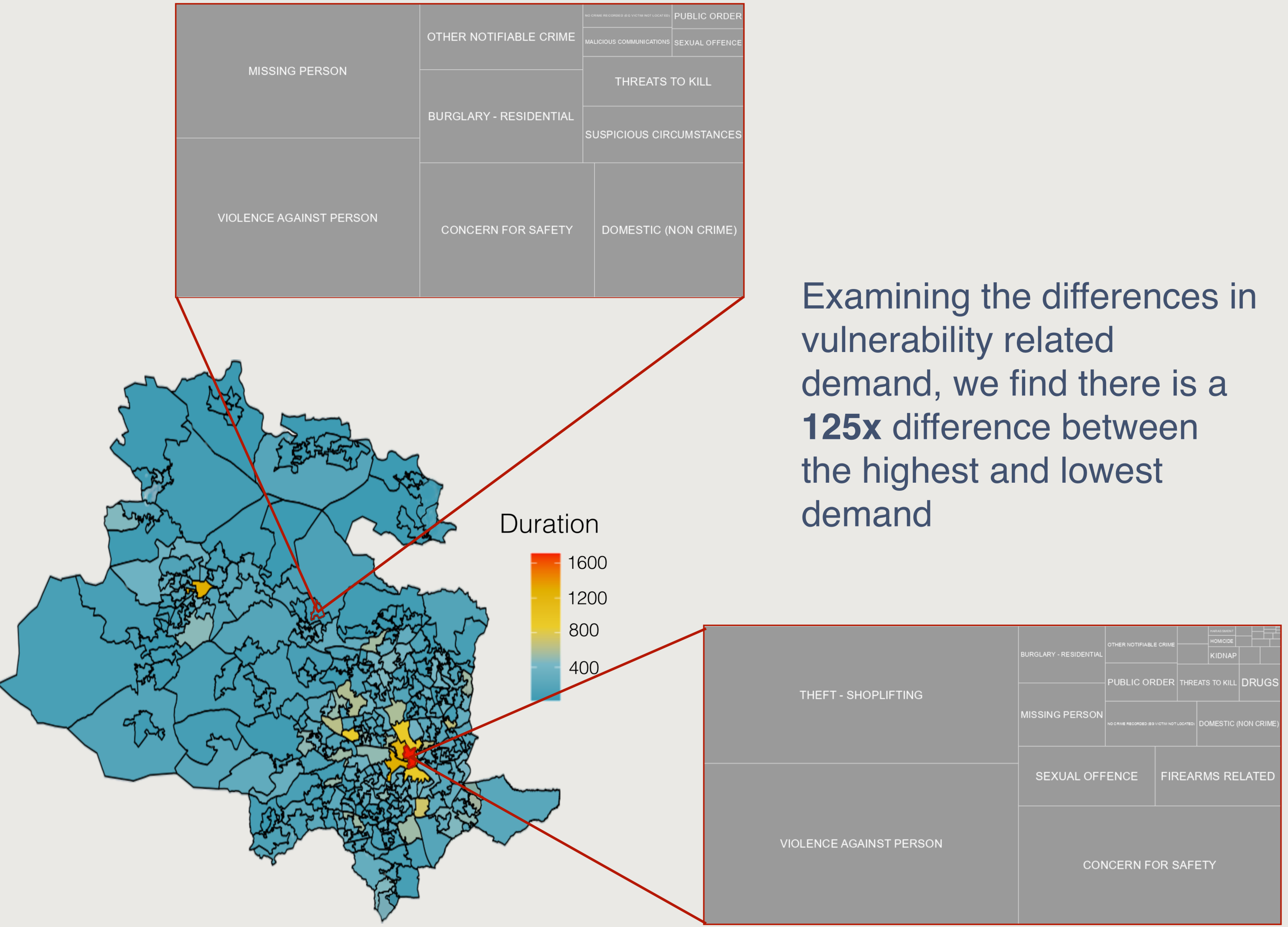


Figure 3 Spatial distribution of duration density across the Bradford Region. Including attended incidents graded emergency and priority. It is also displaying proportional areas charts, showing the proportion of types making up the demands.

Further Research

- Better understand the profiles of problems areas experience to better inform who is best placed to respond**
There is considerable variation in the nature and scale of demand across areas. Understanding how and why, would lead to better understand how demand arises (or dissapears).
- Investigate more effective methods to identify specific problems within routinely collected data.**
Police and their partners record unstructured data regharding the incidents they respond to in incident logs. Anonymised versions of these logs can be analysed using natural language methods - these approaches may provide more effective means to generate detailed understanding of the problems communities face.

This study is based in part on data from Connected Bradford (REC 18/YH/0200 & 22/EM/0127). The data is provided by the citizens of Bradford and district, and collected by the NHS, DfE and other organisations as part of their care and support. The interpretation and conclusions contained in this study are those of the authors alone. The NHS, DfE and other organisations do not accept responsibility for inferences and conclusions derived from their data by third parties